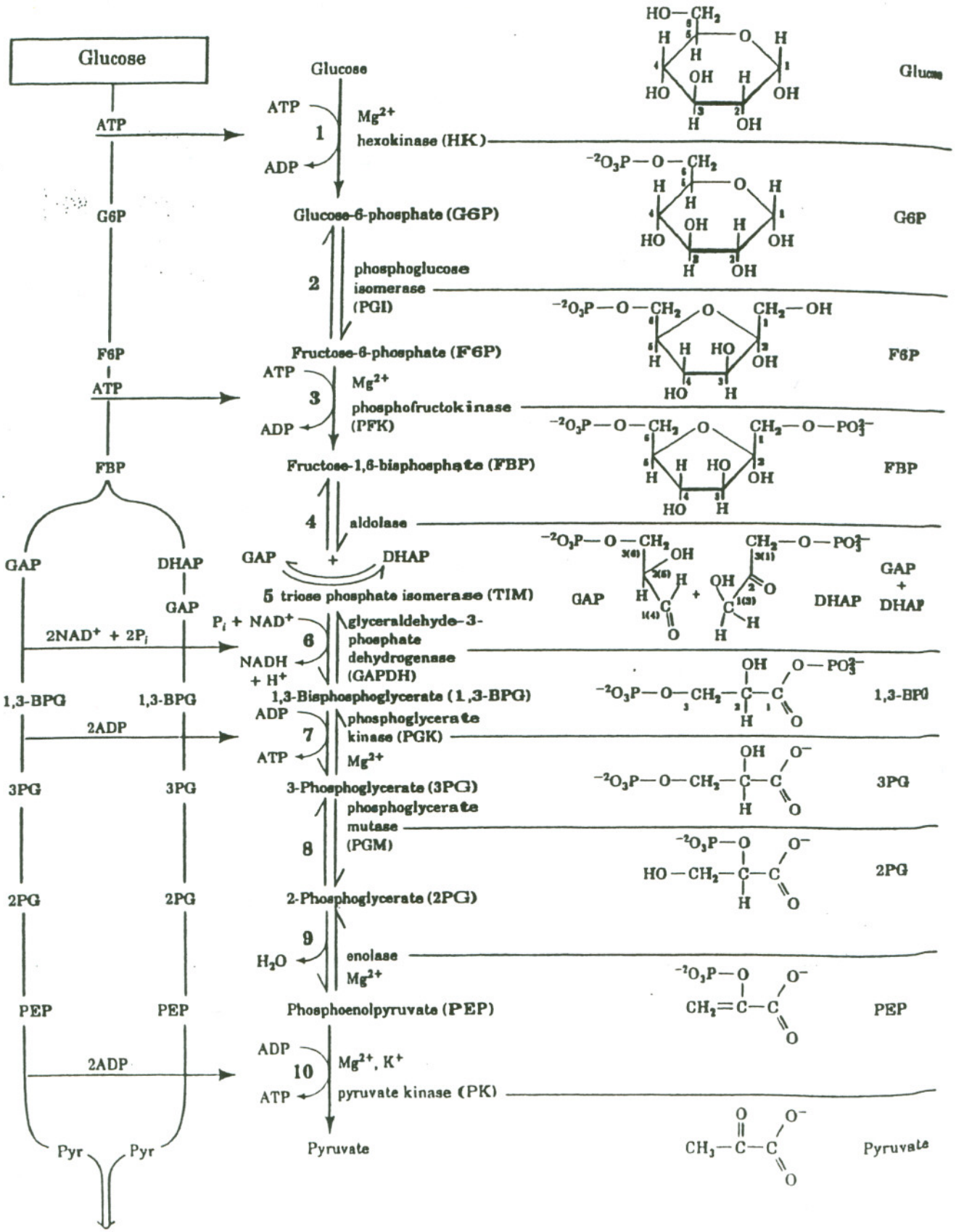
Biochemistry Cumulative Exam

(Supplemental Information Attached)

A mouse is fed a diet including the amino acid glutamate, but no carbohydrate or fatty acids. Small amounts of the essential amino acids and the necessary vitamins and minerals are also included in the diet.

1. (60) Can the mouse synthesize glucose on this diet using glutamate as the sole source of carbon and energy? If the answer is yes, then outline a metabolic pathway for this process. How many glutamate molecules must be used to provide enough carbon atoms, ATP, and NADH molecules to synthesize each molecule of glucose? Write a balanced equation for this process giving the numbers of ATP and CO_2 generated and the number of O_2 consumed.

2. (40) If glutamate radiolabeled with ^{14}C at carbon number 3 is fed to a mouse under the conditions of question 1, which carbon atom(s) of phosphoenolpyruvate, glyceraldehyde-3-phosphate, and glucose would first be labeled with ^{14}C?


Glucose
ATP
G6P
F6P
ATP
FBP
GAP
DHAP
GAP
2NAD+ + 2Pi
1,3-BPG
1,3-BPG
2ADP
3PG
3PG
2PG
2PG
PEP
PEP
2ADP
Pyr
Pyr
Glucose
ATP
1
Mg2+
hexokinase (HK)
ADP
Glucose-6-phosphate (G6P)
2
phosphoglucose isomerase (PGI)
Fructose-6-phosphate (F6P)
ATP
3
Mg2+
phosphofructokinase (PFK)
ADP
Fructose-1,6-bisphosphate (FBP)
4
aldolase
GAP
DHAP
5 triose phosphate isomerase (TIM)
Pi + NAD+
6
glyceraldehyde-3-phosphate dehydrogenase (GAPDH)
NADH + H+
1,3-Bisphosphoglycerate (1,3-BPG)
ADP
7
phosphoglycerate kinase (PGK)
ATP
Mg2+
3-Phosphoglycerate (3PG)
8
phosphoglycerate mutase (PGM)
2-Phosphoglycerate (2PG)
9
H2O
enolase
Mg2+
Phosphoenolpyruvate (PEP)
ADP
10
Mg2+, K+
pyruvate kinase (PK)
ATP
Pyruvate
Glucose
G6P
F6P
FBP
GAP
DHAP
GAP + DHAP
1,3-BPG
3PG
2PG
PEP
Pyruvate

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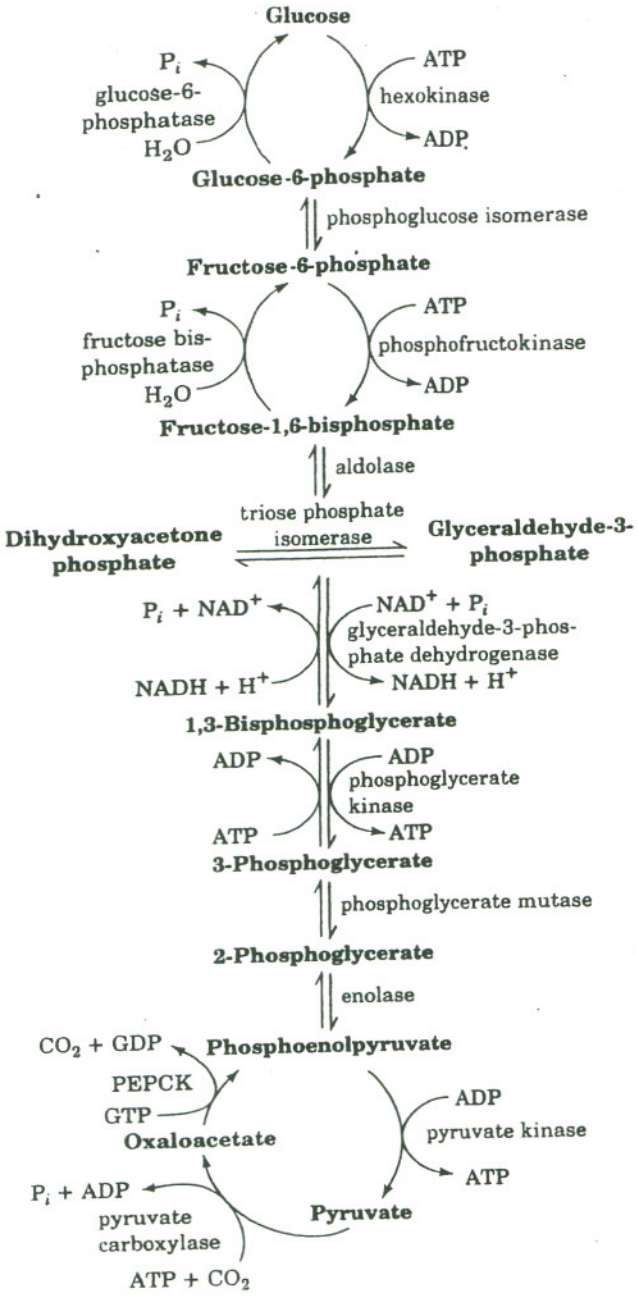
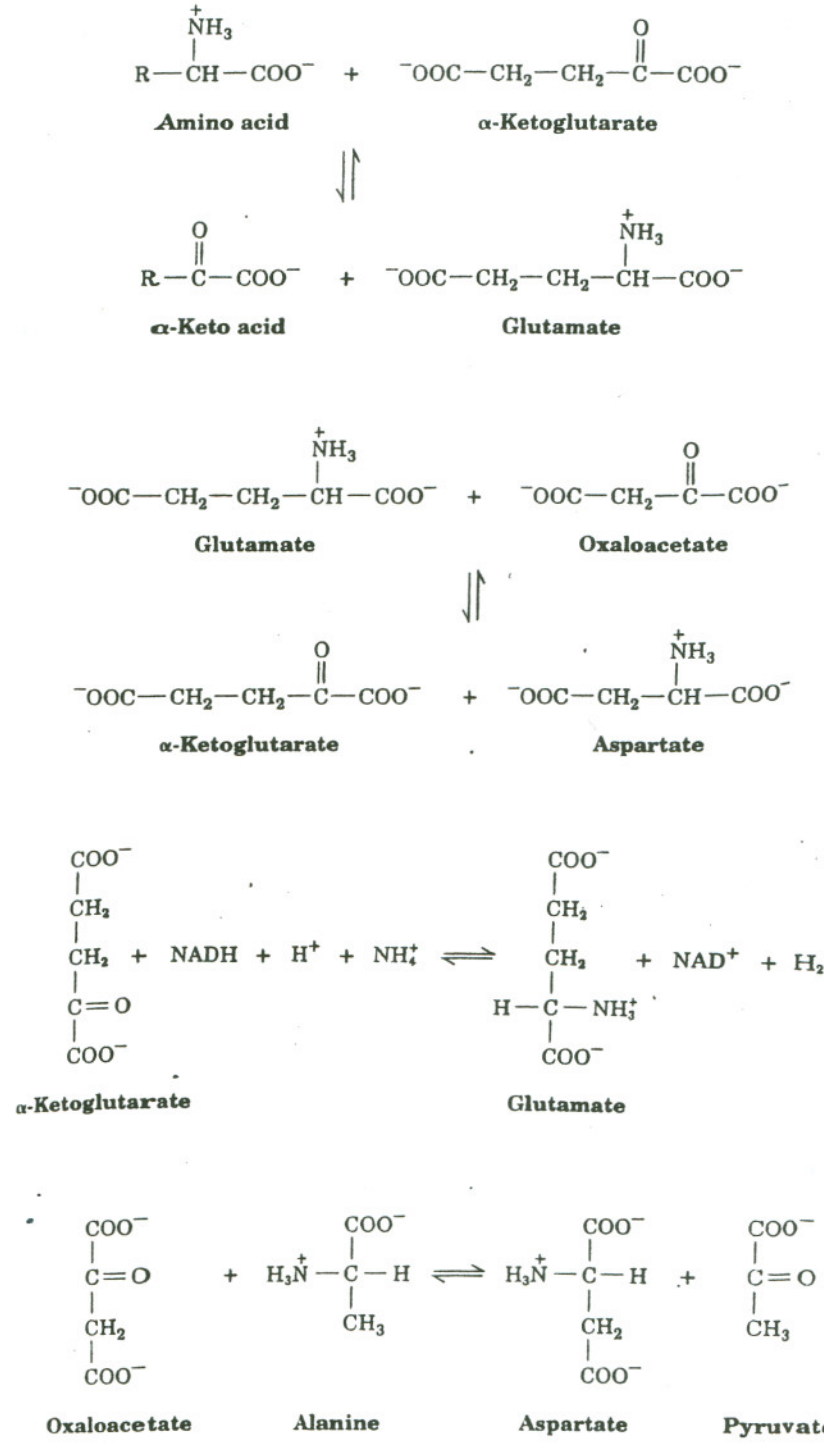
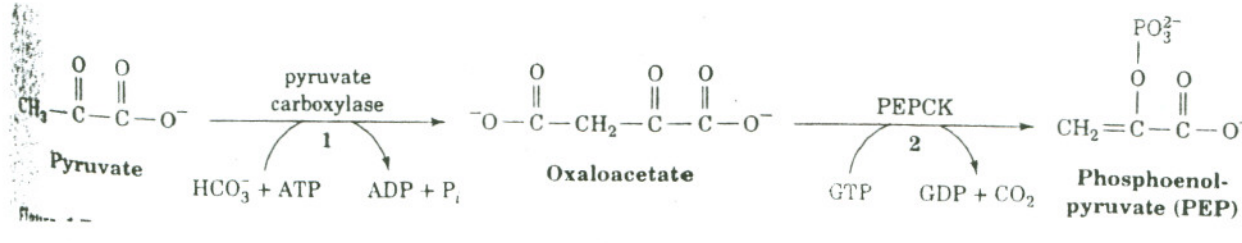


Figure 15-23 ***Key to Metabolism.*** **Comparison of the pathways of gluconeogenesis and glycolysis.** The red arrows represent the steps that are catalyzed by different enzymes in gluconeogenesis. The other seven reaction steps of gluconeogenesis are catalyzed by glycolytic enzymes that function near equilibrium. **See the Animated Figures.**





B-3

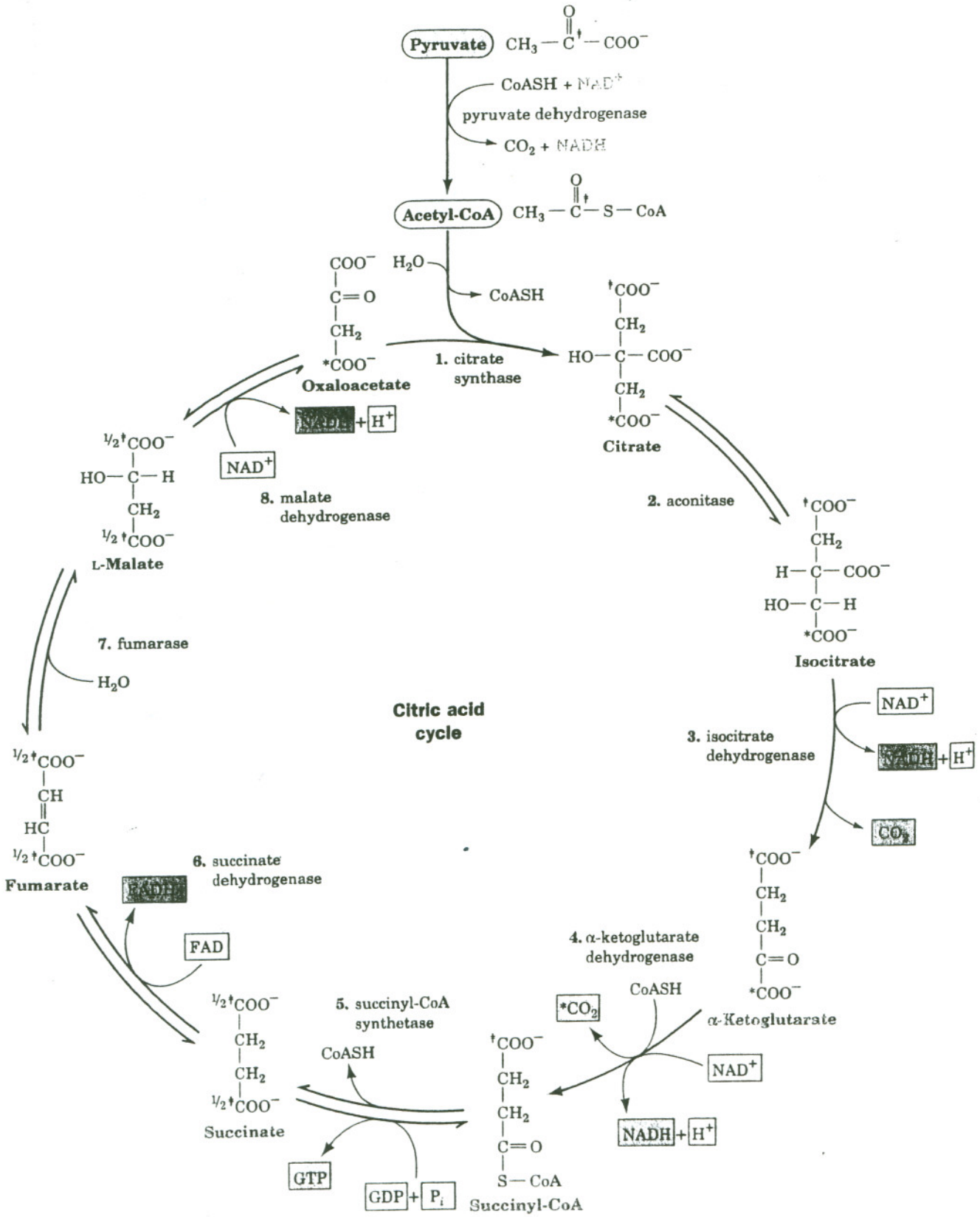


Figure 16-2 ***Key to Metabolism.*** **The reactions of the citric acid cycle.** The reactants and products of this catalytic cycle are boxed. The pyruvate → acetyl-CoA reaction (*top*) supplies the cycle's substrate via carbohydrate metabolism but is not considered to be part of the cycle. An isotopic label at C4 of oxaloacetate (*) becomes C1 of α-ketoglutarate and is released as CO_2 in Reaction 4. An isotopic label at C1 of acetyl-CoA (†) becomes C5 of α-ketoglutarate and is scrambled in Reaction 5 between C1 and C4 of succinate (½†). **See the Animated Figures.**

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